

A sharp two-traversal lower bound for one-sided singular damping

Candidate partial answer to the second open problem in arXiv:2002.03440

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Status: candidate partial result, likely valid. For the Dirichlet damped wave equation on the unit interval, no damping that is regular away from at most one endpoint can make every energy solution extinct before time 2. Thus the extinction time 2 of the critical damping $a(x) = 1/x$ is optimal throughout this one-sided-singularity class. The unrestricted problem remains open when critical nonintegrable damping is allowed at both endpoints or in the interior.

1 Source problem

Freitas, Hefti, and Siegl study

$$u_{tt} + 2a(x)u_t = u_{xx}, \quad 0 < x < 1, \quad u(0, t) = u(1, t) = 0, \quad (1)$$

on the energy space $\mathcal{H} = H_0^1(0, 1) \times L^2(0, 1)$ [1]. For $a(x) = 1/x$, every solution vanishes after time 2. At the end of Section 4 (arXiv PDF page 10), they ask:

“Are there damping terms which will cause finite time extinction to occur for all solutions for all time larger than some T which is now smaller than 2 (on an interval with unit length)?”

For a strongly continuous solution semigroup $S_a(t)$, saying that every solution is zero for all $t > T$ is equivalent to $S_a(T) = 0$: strong continuity supplies the endpoint once $S_a(t) = 0$ for $t > T$.

2 Result and scope

Call a *one-sided admissible* if

$$a \in C^1((0, 1]), \quad a \geq 0, \quad \sup_{0 < x \leq 1} xa(x) < \infty, \quad (2)$$

and use the standard maximal-dissipative (form) realization of (1) on $\mathcal{H}$. The Hardy estimate gives the same energy realization used for α/x in [1]; the only fact about this realization needed below is ordinary local finite propagation on every strip $[\delta, 1] \times [0, T]$, where a is bounded and smooth. The theorem therefore also applies to any broader energy realization that agrees locally with the classical equation away from $x = 0$.

for some L^p norm and a given positive number a_0 , for instance. We then want to know if an optimal damping exists and, if so, what is its shape. The answer to these questions will be, in general, expected to depend on the value of a_0 .

If, on the other hand, we are interested in the decay rate of solutions of (1) without imposing such a restriction, the natural question becomes whether or not it is possible to improve upon the extinction time, that is, the time after which all solutions vanish. More precisely, are there damping terms which will cause finite time extinction to occur for all solutions for all time larger than some T which is now smaller than 2 (on an interval with unit length)?

Figure 1: The second open problem in Section 4 of arXiv:2002.03440, PDF page 10.

Theorem 1 (Two-traversal lower bound). *If a is one-sided admissible, then*

$$S_a(T) \neq 0 \quad (0 < T < 2).$$

Consequently, a universal finite extinction time in this class is at least 2.

The reflected statement holds when a may be singular only at $x = 1$. The growth bound in (2) is a convenient sufficient condition for the energy realization, not the mechanism behind the lower bound. The proof only uses that a is integrable along the broken characteristic before it reaches the singular endpoint.

Corollary 2 (Sharpness of the known example). *For $a(x) = 1/x$, the universal extinction time is exactly 2.*

Indeed, Theorem 1 gives the lower bound, while the extinction result recalled and reproved in [1] gives the upper bound.

3 Proof intuition

Write the wave as two first-order travelling components. Put

$$p = u_t + u_x, \quad q = u_t - u_x.$$

Then p travels left, q travels right, and damping couples them without changing their characteristic speeds. Start with energy data for which q has one jump at a point $x_0 > 0$ and p is continuous. Along the right-going ray, the jump is multiplied by an exponential of a finite damping integral, so it cannot disappear. At the regular endpoint $x = 1$, the Dirichlet condition reflects it into a nonzero jump of p . That jump then travels left and remains nonzero until it reaches the only possible singularity at $x = 0$, at time $2 - x_0$.

Given any $T < 2$, choose $x_0 < 2 - T$. The reflected jump is still in the interior at time T , so the state cannot be zero. The key point is not an energy estimate: it is the impossibility of annihilating a wavefront by a finite damping integral.

4 Jump propagation

The following elementary lemma records the local mechanism. All jumps are differences of the two one-sided traces.

Lemma 3 (Characteristic jump law). *On a region where a is locally integrable and a solution of (1) is piecewise C^1 , the Riemann invariants satisfy*

$$p_t - p_x = -a(p + q), \quad q_t + q_x = -a(p + q). \quad (3)$$

If initially q has a jump J at x_0 and p is continuous there, then, until the ray $x = x_0 + t$ meets the boundary,

$$[p] = 0, \quad [q](t) = J \exp\left(-\int_0^t a(x_0 + s) ds\right). \quad (4)$$

At $x = 1$ the jump reflects into p with the opposite sign. On the outgoing ray $x = 2 - x_0 - t$, its amplitude is

$$[p](t) = -J \exp\left(-\int_{x_0}^1 a(s) ds - \int_{2-x_0-t}^1 a(s) ds\right), \quad 1 - x_0 < t < 2 - x_0. \quad (5)$$

In particular, every displayed amplitude is nonzero when the integrals are finite.

Proof. Equation (3) follows directly from $u_t = (p + q)/2$ and $u_x = (p - q)/2$. Across the right-going characteristic $x - x_0 - t = 0$, a jump of p would contribute a Dirac mass with coefficient $-2[p]$ to $p_t - p_x$. The right side of (3) has no Dirac mass, so $[p] = 0$. Taking ordinary trace differences in the equation for q now gives

$$\frac{d}{dt}[q](t) = -a(x_0 + t)[q](t),$$

which proves (4).

For a piecewise classical Dirichlet solution, $u(1, t) = 0$ implies $u_t(1, t) = 0$ and hence $p(1, t) + q(1, t) = 0$. The time jump arriving in q therefore creates the opposite time jump in the boundary value of p . Repeating the preceding calculation on the left-going characteristic gives the second exponential in (5). This jump calculus is local; standard characteristic approximation extends it to the corresponding piecewise-smooth energy solution. $\square$

5 Proof of Theorem 1

Fix $0 < T < 2$. Choose

$$0 < x_0 < \min\{1, 2 - T\}, \quad x_0 \neq 1 - T. \quad (6)$$

The excluded value merely prevents the observed time from being the instant when the jump is at the boundary.

We first construct globally legitimate energy data with exactly the desired local jump. Choose a smooth function χ on $[x_0, 1]$ which equals 1 near x_0 and vanishes near 1. Next choose a unit-integral function

$$\psi \in C_c^\infty((x_0/4, x_0/2)), \quad \int_0^1 \psi = 1,$$

and put

$$c = \int_{x_0}^1 \chi(s) ds, \quad q_0(x) = \mathbf{1}_{(x_0, 1)}(x)\chi(x) - c\psi(x), \quad p_0(x) = 0.$$

Then q_0 has a single jump of size 1 at x_0 , is smooth elsewhere, and has integral zero. Define

$$u_0(x) = -\frac{1}{2} \int_0^x q_0(s) ds, \quad u_1(x) = \frac{1}{2} q_0(x). \quad (7)$$

Thus $u_0 \in H_0^1(0, 1)$, $u_1 \in L^2(0, 1)$, and the corresponding initial invariants are precisely $p_0 = 0$ and q_0 .

The broken ray issued from the jump is

$$\gamma(t) = \begin{cases} x_0 + t, & 0 \leq t \leq 1 - x_0, \\ 2 - x_0 - t, & 1 - x_0 \leq t < 2 - x_0. \end{cases} \quad (8)$$

By (6), $T < 2 - x_0$ and $\gamma(T) > 0$; moreover $\gamma(T) \neq 1$. The entire ray segment up to time T lies in $[\delta, 1]$ for some $\delta > 0$. Hence a is bounded there, both damping integrals in Lemma 3 are finite, and the appropriate invariant has a nonzero jump at the interior point $\gamma(T)$. Therefore

$$S_a(T)(u_0, u_1) \neq (0, 0).$$

Since $T < 2$ was arbitrary, the theorem follows. $\square$

Remark 4 (The critical example). *For $a(x) = 1/x$, the incoming amplitude in (4) is $Jx_0/(x_0 + t)$. After reflection, (5) becomes*

$$[p](t) = -Jx_0(2 - x_0 - t),$$

which is nonzero until the ray reaches $x = 0$ at time $2 - x_0$. Letting $x_0 \downarrow 0$ displays geometrically why the sharp universal threshold is 2 even though each fixed tracked jump dies slightly earlier.

6 Focused upgrade attempts

Because the characteristic theorem is a substantial partial answer, eight focused attempts were made to upgrade it to the unrestricted question.

No.	Route	Outcome
1	Isolated wavefront	Gave the nonvanishing exponential jump law.
2	Energy-compatible data	The correction in (7) produces a genuine $H_0^1 \times L^2$ state without adding another jump.
3	Boundary reflection	Extended survival from one crossing to the full broken-ray time $2 - x_0$.
4	Rough coefficients	The proof only needs a finite line integral along the ray; the packet states a conservative smooth-away-from-zero class to keep the realization unambiguous.
5	Two endpoint poles	The candidate $a = 1/x + 1/(1 - x)$ defeats this ray: the reflection point itself has infinite damping. Its spectral ODE did not yield a nilpotent semigroup or a $T < 2$ formula.
6	Spectral determinant	Nilpotence forces empty spectrum, but classifying zero-free entire Dirichlet determinants for arbitrary critical singularities is an unresolved new problem.
7	Darboux/gauge factorization	A single exponential-affine factorization forces a one-pole logarithmic derivative $a = \pm h'/h$; it does not cover two poles or interior barriers.
8	Interior absorbers and literature threshold	A critical interior singularity requires a nontrivial choice of energy trace/interface realization. Existing backward-uniqueness theorems exclude broad subcritical classes but do not settle this critical regime.

The remaining full problem is therefore concentrated in damping with a nonintegrable critical barrier at both endpoints or in the interior. Neither the characteristic proof nor the located literature classifies the resulting singular scattering realization. Continuing without such a new theorem had low probability of producing a rigorous full answer.

7 Verification and novelty status

The proof was audited at the following points: the factor of a in (3); the distributional coefficient forcing $[p] = 0$; the reflection sign; both exponential integrals; the zero-integral correction in the initial data; the strict inequalities in (6); and the exact $a = 1/x$ amplitude. The accompanying script checks the broken-ray geometry and the explicit critical amplitude on a grid of 399 test times. Those computations are sanity checks, not part of the proof.

A bounded novelty search on 13 August 2026 covered the local run indexes, the exact source title and quoted question, “singular damping extinction time smaller than 2,” “minimal extinction time,” the arXiv record through 2026, and later work citing the critical example. Kleinhenz and Wang prove optimal abstract backward uniqueness under a subcritical unboundedness threshold in arXiv:2310.19911 [2]; their later Theorem 4 rules out finite-time extinction for normally L^p damping, $p > 1$ [3]. Both explicitly contrast their result with the critical $2/x$ example. Neither gives a minimal-time theorem in the critical one-sided class or answers the two-sided/interior-singular case. No later answer to the exact $T < 2$ question was found. Novelty is plausible but not certified, and priority is not claimed; the argument may be implicit in standard one-dimensional propagation theory.

8 Limitations and review recommendation

This is not a full answer to either open problem in [1]. It does not address the norm-constrained optimization question. For the second problem, it does not rule out a damping with critical nonintegrable singularities at both endpoints or at an interior interface, and it does not construct such a damping. It also does not claim a general semigroup construction beyond the standard one-sided form class (2).

Human review should focus first on the local-to-global use of the piecewise-smooth energy solution in Lemma 3, then on the boundary reflection step. Once those standard hyperbolic facts are accepted, every remaining step is explicit. The best route to a full result would be a classification of maximal-dissipative interface conditions for critical interior/two-endpoint singularities, followed by a zero-free spectral- determinant or characteristic-scattering analysis.

References

- [1] P. Freitas, N. Hefti, and P. Siegl, *The damped wave equation with singular damping*, Proc. Amer. Math. Soc. **148** (2020), 4273–4284; arXiv:2002.03440.
- [2] P. Kleinhenz and R. P. T. Wang, *Optimal backward uniqueness and polynomial stability of second order equations with unbounded damping*, arXiv:2310.19911 (2023).
- [3] P. Kleinhenz and R. P. T. Wang, *Sharp polynomial decay for polynomially singular damping on the torus*, Ann. PDE **12** (2026), Article 6, doi:10.1007/s40818-025-00230-2.